MIL-STD-262
11 March 1957

# MILITARY STANDARD

# TEST AND INSPECTION OF CRANE TRUCK, WAREHOUSE, ELECTRIC SOLID RUBBER TIRES

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# OFFICE OF THE ASSISTANT SECRETARY OF DEFENSE
# WASHINGTON 25, D. C.

Supply and Logistics

# TEST AND INSPECTION
# OF
# CRANE TRUCK, WAREHOUSE, ELECTRIC, SOLID RUBBER TIRES

MIL–STD–262 11 March 1957

1. This standard has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force, effective 15 April 1957.

2. In accordance with established procedure, the Standardization Division has designated the Quartermaster Corps, the Bureau of Supplies and Accounts, and the Air Force, respectively, as Army-Navy-Air Force custodians of this standard.

3. Recommended corrections, additions, or deletions should be addressed to the Standardization Division, Office of the Assistant Secretary of Defense (Supply and Logistics), Washington 25, D. C.



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## LIST OF DRAWINGS

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M 6,000 Pound Standard Crane Test Load
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P Details of load positions 1, 2, and 3
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## TEST FORMS

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5G Towing Hooks
7G Speed
8G Slope Ascension
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16G Acceleration
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# TEST AND INSPECTION
# OF
# CRANE TRUCK, WAREHOUSE, ELECTRIC, SOLID RUBBER TIRES

## 1. SCOPE

**1.1 Scope.** This standard covers the testing and inspection of equipment as listed in Specification MIL-C-3988.

## 2. REFERENCED DOCUMENTS

**2.1 Specifications.** The following specifications and publications of the issue in effect on date of invitation for bids, form a part of this standard.

### SPECIFICATIONS

MILITARY

MIL-C-3988 — Crane Truck, Warehouse, Electric, Solid Rubber Tires.

### STANDARDS

MILITARY

MS-15367 — Batteries, Storage for Electric Operated Industrial Trucks and Tractors.

### PUBLICATIONS

MILITARY

Joint Storage and Materials Handling Manual

TM-743-200 — Department of the Army Technical Manual
NAVSANDA Publication 284 — Department of the Navy
AFM 67-3 — Department of the Air Force
NAVMC 1101 — U. S. Marine Corps. Joint Publication.
NAVSANDA Publication 249 — Storage and Materials Handling Handbook

(Copies of specifications and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

## 3. DEFINITIONS. Not applicable.

## 4. GENERAL REQUIREMENTS

**4.1 Requirements for approval.** Items of equipment tested and inspected under this standard shall pass the following tests and inspections outlined herein:

A. Pre-Test

B. Performance Test

C. Post-Test

**4.2 Equipment operators.**

4.2.1 *Equipment operators.* Equipment operators shall be furnished by the government to assist the Test Officer in conducting the test and inspections outlined in this standard.

4.2.2 *Operator qualification.* Operators assigned for testing the equipment under the provisions of this standard shall have completed satisfactorily the applicable portions of the course entitled "On-The-Job Training Course for Operators of Materials Handling Equipment", Joint Storage and Materials Handling Manual. He shall have available to show to the contractor's representatives if requested, his "Standard Operator's Permit" (DDForm 313), or its equivalent, for the type of equipment under test.



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4.2.3 *Operator indoctrination.* Operators provided for testing any machine under provisions of this standard shall be required to drive the machine through the appropriate test course until he has achieved sufficient proficiency in operating the machine. This warmup period shall last for no more than two hours per driver. During it, the contractor's representative will be permitted to inform each driver of the proper technique to be used in operating the machine to achieve maximum performance.

**4.3 Test location.** The test location will be as designated by the Contracting Officer.

**4.4 Test officer.** The test officer shall be appointed by the Department of Defense and shall exercise full authority in directing the test.

**4.5 Ambient temperature.** Temperature of the surrounding atmosphere may range from 40° to 104°F. inclusive. If the temperature drops below or rises above this range the test will be discontinued until the temperature returns to this range.

**4.6 Contractor's representatives.** Any contractor whose machine is being tested in accordance with the provisions of this standard will be permitted to have present representatives to witness the test and inspection and assist the Test Officer (upon request).

**4.7 Official test report.** An official report of the test and inspection will be prepared. It will include records of the pre-test, performance tests, post-test and such other tests as were deemed necessary together with a statement from the Test Officer indicating whether the machine has passed or failed. The Test Officer will prepare as many copies as directed by the Contracting Officer.

## 5. DETAILED REQUIREMENTS

**5.1 Pre-Test.**

5.1.1 *Description.* The pre-test shall consist of three principal parts, each of which is described below:

A. Preventive Maintenance Service and Inspection

B. Inspection Check List

C. Inspection Tests

5.1.2 *Standard procedures.*

5.1.2.1 *Standard load specifications.* Inspections and tests under the provisions of this standard requiring use of standard loads will utilize the appropriate load detailed in Drawings L, M, and N or a load of another construction or composition which has the same dimensions, weight, location of center of gravity, and safety features, as the applicable load detailed in Drawings L, M, and N. A tolerance of $-1\%$ $+$ $0\%$ of the specified test loads will be permitted.

5.1.2.2 *Pre-test procedure.* All items of the pre-test shall be completed satisfactorily in turn, prior to the performance test.

5.1.3 *Preventive maintenance service and inspections procedures.* Preventive Maintenance Service and Inspections ("B" Schedule) shall be conducted as contained in the Storage and Materials Handling Handbook, NAVSANDA Publication 249.

5.1.4 *Check list.*

5.1.4.1 *When performed.* All inspections and other procedures needed to answer the questions contained in the Pre-test Check List shall be performed in conjunction with, or immediately subsequent to, the preventive maintenance service and inspection.

5.1.4.2 *How recorded.* Each of the questions contained in the Pre-test Check List shall be answered by personnel designated by the Test Officer by placing an "X" in the column headed "Yes" or "No" as appropriate.



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5.1.4.3 *How evaluated.* After completion of the check list, any "X" in the "No" column shall be cause for rejection of the machine under test.

5.1.4.4 *Other checks, type EE trucks.* In the case of type EE trucks, in addition to all other inspections and tests described herein, requirements outlined in the following paragraphs of Specification MIL–C–3988 shall be inspected or tested for compliance.

| *Specification Paragraph No.* | *Component* |
|---|---|
| 3.1.5.2.2 | Enclosures and shields. |
| 3.1.5.4 | Resistor housings and heat-actuated devices. |
| 3.2.1.1 | Motors. |
| 3.4 | Controls and instrumentation. |
| 3.5 | Accessories. |

5.1.5 *Pre-Test Inspection tests.* Inspection tests shall be given in accordance with the following tabulation, to indicate fulfillment of the referenced paragraphs of Specification MIL–C–3988, as provided for in test forms.

5.1.5.1 *Specific inspection tests.*

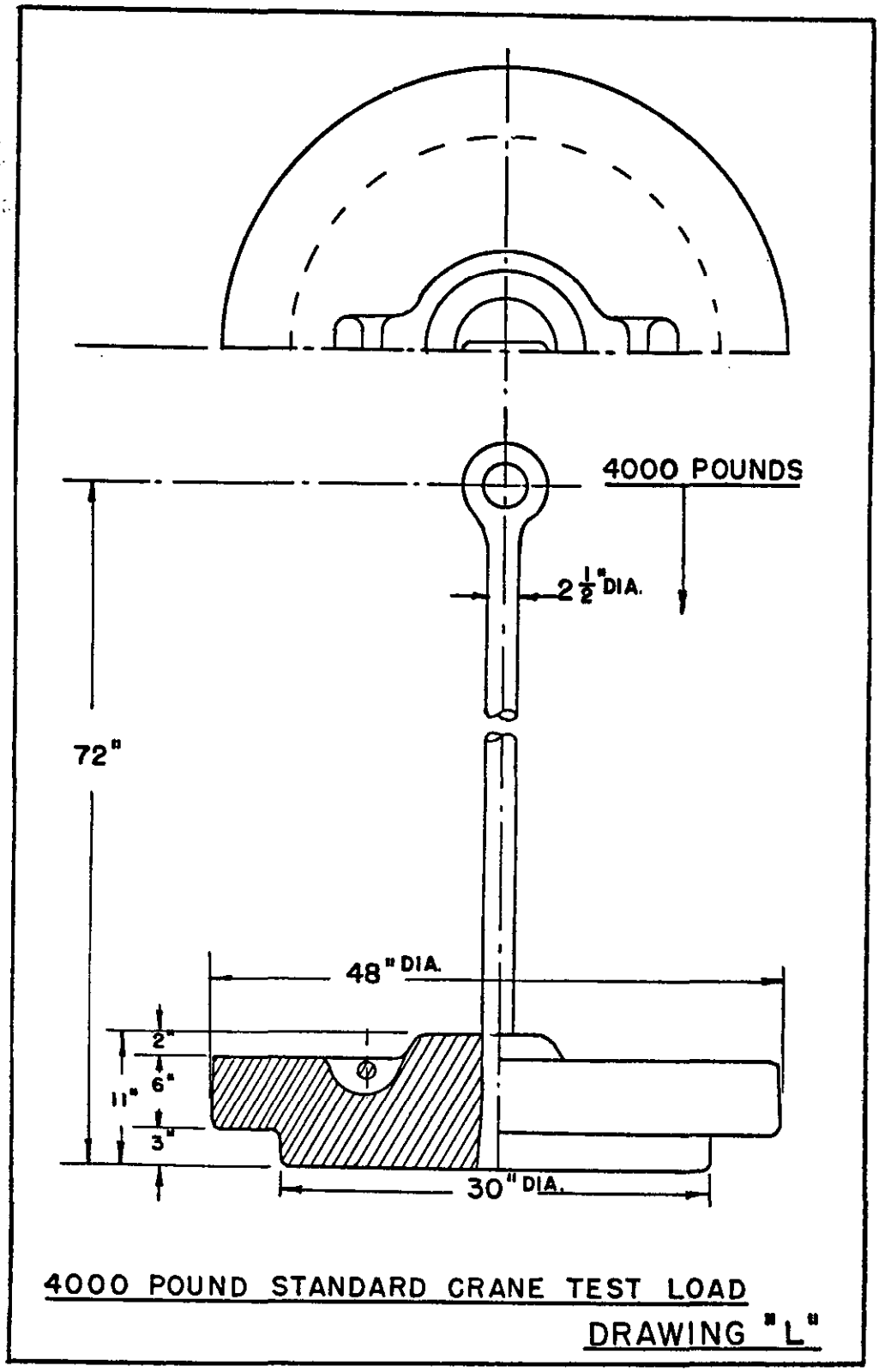
| *Applicable Test Form* | *Test Title* | *Applicable Paragraph of Specification* |
|---|---|---|
| 2G | Steering Wheel | 3.6.7 |
| 5G | Towing Hooks | 3.3.5.3 |
| 7G | Speed | 3.6.4 |
| 8G | Slope Ascension | 3.6.5 |
| 14G | Stability | 3.6.13 |
| 15G | Wear of Steering Mechanism | 3.6.7 |
| 16G | Acceleration | 3.6.14 |
| 17G | Stopping Distance | 3.6.15 |

5.1.5.2 *How recorded.* Each of the questions contained in the Pre-Test Inspection Check Sheet shall be answered by personnel designated by the Test Officer by placing an "X" in the column headed "Yes" or "No" as appropriate.

5.1.5.3 *How evaluated.* After completion of these additional inspections, all test forms shall be examined. Any "X" in the "No" column of 1 or more of these forms shall be cause for rejection of the machine under test.



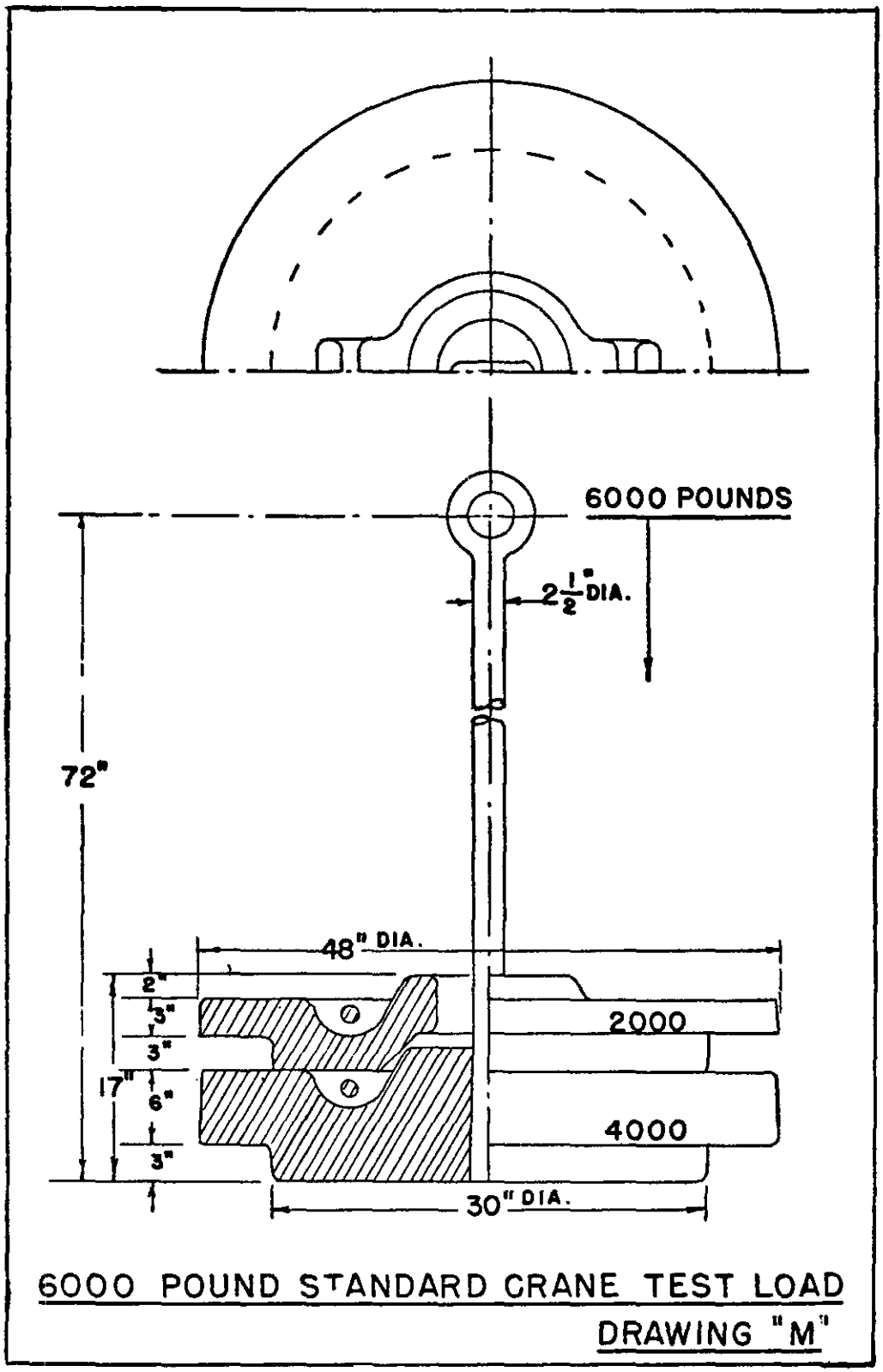
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4000 POUND STANDARD CRANE TEST LOAD
DRAWING "L"



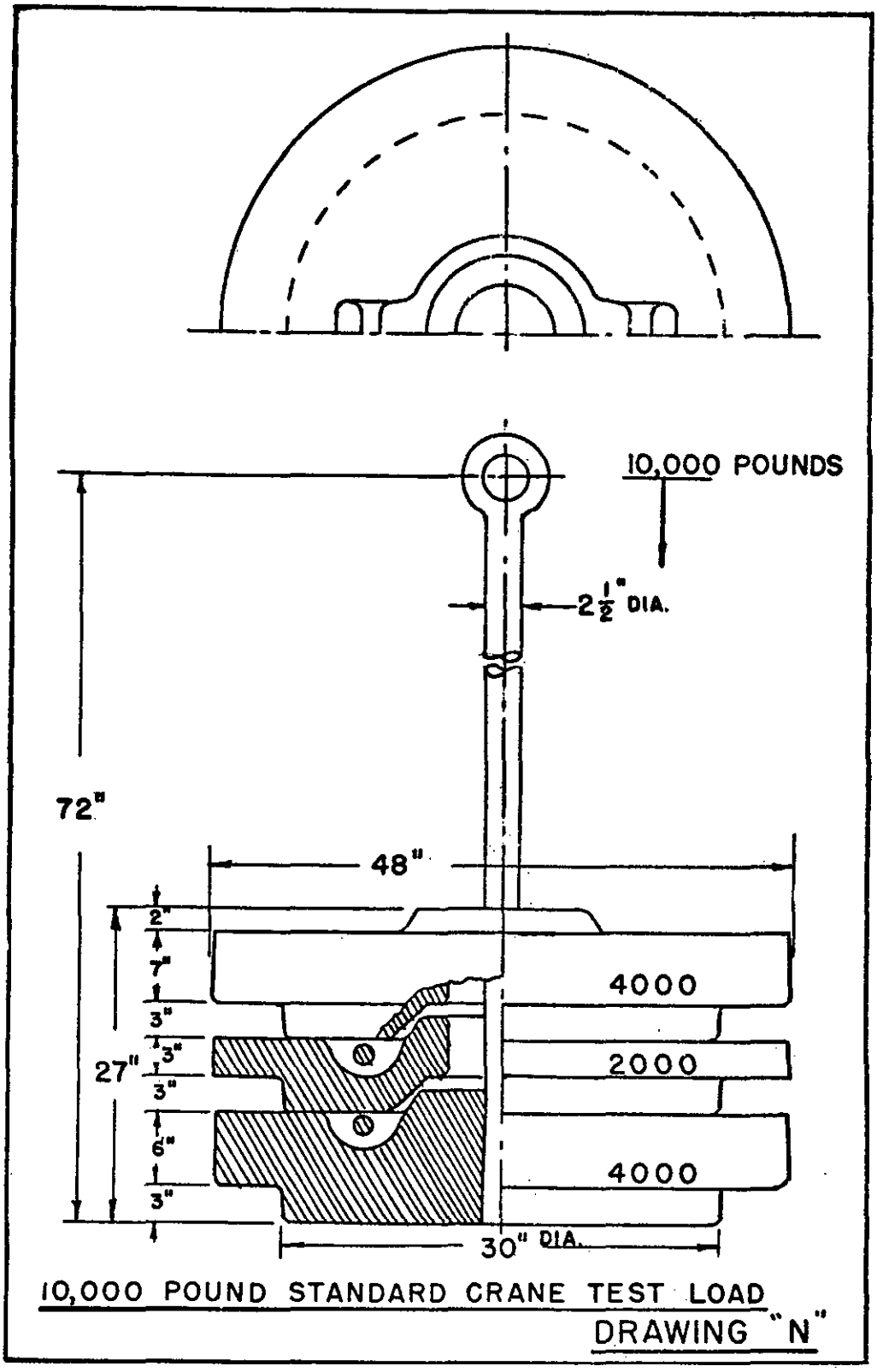
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6000 POUND STANDARD CRANE TEST LOAD
DRAWING "M"



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10,000 POUND STANDARD CRANE TEST LOAD
DRAWING "N"



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# PRE-TEST INSPECTION CHECK LIST
# CRANE TRUCK, WAREHOUSE, ELECTRIC, SOLID RUBBER TIRES

Description of Truck_____

Truck Mfgr._____ Date_____

Model_____ Test Location_____

Serial No._____ Recorded by_____

| *Component* | *Applicable Specification Paragraph* | *Item To Be Checked* | *Yes* | *No* |
|---|---|---|---|---|
| Insulation | 3.1.5.1 | Has contractor submitted to the inspection officer a certificate of compliance with the requirements of this paragraph? | | |
| Enclosures and shields | 3.1.5.2.1 | If type E truck is specified, has contractor submitted to the inspection officer a certificate of compliance with the requirements of this paragraph? | | |
| | 3.1.5.2.2 | If type EE truck is specified, has contractor submitted to the inspection officer a certificate of compliance with the requirements of this paragraph? | | |
| Overcurrent devices | 3.1.5.3 | Has contractor submitted to the inspection officer a certificate of compliance with the requirements of this paragraph? | | |
| Resistor housings and heat-actuated devices | 3.1.5.4 | Has contractor submitted to the inspection officer a certificate of compliance with the requirements of this paragraph? | | |
| Static discharge device | 3.1.5.5 | If type EE truck is specified, is it equipped with two replaceable electrically conductive rubber straps mounted adjacent to or immediately inside drive wheels to discharge static electricity? | | |
| Control panel | 3.1.5.6 | Does the control panel comply with the requirements of this paragraph? | | |
| Motors | 3.2.1.1 | Has contractor submitted to the inspection officer a certificate of compliance with the requirements of this paragraph? | | |
| Speed controller | 3.2.1.2 | Does speed controller provide for at least 4 speeds forward? | | |
| | 3.2.1.2 | Does speed controller provide for at least 4 speeds reverse? | | |
| | 3.2.1.2.1 | Does the controller have automatic means to prevent reversing except through first speed? | | |
| | 3.2.1.2 | Does speed controller have automatic means to prevent starting or reversing except in first speed? | | |
| | 3.2.1.2 | Does controller permit precision maneuvering in restricted areas? | | |
| Lead line hoist, boom hoist, and boom sluing controllers | 3.2.1.2.2 | When specified, do load line hoist, boom hoist, and boom sluing controllers provide for at least 2 speeds in either direction through a stepping mechanism? | | |



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| *Component* | *Applicable Specification Paragraph* | *Item To Be Checked* | *Yes* | *No* |
|---|---|---|---|---|
| Battery | 3.2.2.1 | If battery(s) is furnished by the contractor with the truck, is it in accordance with Standard MS–15367 (36 volt group) and as specified in procurement documents? | | |
| | 3.2.2.1 | If battery(s) is furnished by contractor with truck, is minimum capacity at least 600 ampere-hours for size I and 450 ampere-hours for size II? | | |
| Battery leads | 3.2.2.1.1 | If battery is furnished by the contractor with truck, is battery equipped with flexible leads with locking half of connector attached? | | |
| | 3.2.2.1.1 | If battery is furnished by contractor with truck, do battery leads extend at least 30 inches from the corner of the battery in accordance with Standard MS–15367? | | |
| | 3.2.2.1.1 | Are all conductors passing through metal walls protected by insulating grommets, frame connectors, or insulated passages? | | |
| | 3.2.2.1.1 | If battery is furnished by the contractor with truck, are battery leads, within the battery compartment, separated as far as possible? | | |
| | 3.2.2.1.1 | If battery is furnished by the contractor with truck, are battery leads permanently connected to battery terminal posts? | | |
| | 3.2.2.1.1 | On size II is provision made for connecting batteries in series? | | |
| Battery connectors | 3.2.2.2 | Is there a connector mounting half located to accommodate the connector locking half on battery leads? | | |
| | 3.2.2.2 | Is connector mounting half permanently affixed to the truck? | | |
| | 3.2.2.2 | If battery(s) is furnished by the contractor with truck, is connector locking half provided with reliable means for grasping to permit removal without strain on the leads? | | |
| | 3.2.2.2 | Is the connector locking half compatible with connector mounting half? | | |
| | 3.2.2.2 | For type EE trucks, if connectors are mounted outside the battery compartment, is provision made for locking each connector with a padlock to be furnished by the Government at destination? | | |
| Battery compartment | 3.2.2.3 | Is battery compartment made from non-combustible material? | | |
| | 3.2.2.3 | If metal battery compartment is used, is it at least number 16 USS gauge in thickness? | | |
| | 3.2.2.3 | Is battery compartment flanged not less than ½-inch on all sides, or provided with equivalent reinforcement? | | |
| | 3.2.2.3 | Can battery compartment cover be opened using hand tools? | | |
| | 3.2.2.3 | Is battery compartment constructed to prevent short circuiting? | | |



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| *Component* | *Applicable Specification Paragraph* | *Item To Be Checked* | *Yes* | *No* |
|---|---|---|---|---|
| Battery compartment (continued) | 3.2.2.3 | Is battery compartment amply ventilated? | | |
| | 3.2.2.3 | If type EE truck is specified, is a liner of ample strength and rigidity to prevent short circuiting of battery terminals secured to inside of cover? | | |
| | 3.2.2.3 | Does battery compartment have following minimum inside clear dimensions: | | |

| *Truck Size* | *Battery Compartment Dimensions* | | |
|---|---|---|---|
| | *Length* | *Width* | *Height* |
| I | 32¾ | 40½ | 25½ |
| II | 40 | 63 | 25½ |

| | | | | |
|---|---|---|---|---|
| | 3.2.2.3 | Are effective means provided for securing battery in place? | | |
| | 3.2.2.3 | If type EE truck is specified, has provision been made for locking the compartment with a padlock? | | |
| | 3.2.2.3 | If type EE truck is specified, are battery compartment openings covered with heavy gauge wire mesh, or the equivalent? | | |
| Reduction units | 3.2.3 | Are drive, boom hoist, boom sluing, and load line hoist reduction units connected directly to their respective counterparts? | | |
| | 3.2.3 | Are drive, boom hoist, and load line hoist reduction units totally enclosed? | | |
| | 3.2.3 | Do drive, boom hoist, and load line hoist reduction units operate in a constant oil bath? | | |
| Chassis and frame | 3.3.1 | Do chassis, frame, crane members, and related structure have sufficient depth to protect working parts? | | |
| | 3.3.1 | Are chassis, frame, crane members, and related structure in operating condition with no evidence of permanent deformation? | | |
| Bumpers | 3.3.1.1 | Is the crane truck provided with a front bumper of structural channel and of sufficient size to protect front tires? | | |
| | 3.3.1.1 | Is the rear bumper of sufficient size and placed so as to protect rear springs, tires, and frame? | | |
| Boom support | 3.3.1.2 | Is the crane truck equipped with a mast, or A-frame, and center pin or a structure of similar characteristics? | | |
| | 3.3.1.2 | Are cross members so designed and constructed so as not to obstruct unduly the operator's view directly ahead or of the load hook? | | |
| Boom pivot | 3.3.1.3 | Is the center pin or boom pivot mounted on roller bearings or oil impregnated bronze bearings? | | |
| Load line hoist | 3.3.1.4 | Are guide rollers or other devices provided to prevent slack cables from riding over drum or sheave flanges, or chafing on crane or boom structure? | | |
| Boom | 3.3.1.4.1 | Is the boom of the 2-section, telescopic, 270° sluing type? | | |



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| *Component* | *Applicable Specification Paragraph* | *Item To Be Checked* | *Yes* | *No* |
|---|---|---|---|---|
| Boom (continued) | 3.3.1.4.1 | Can the boom be moved through a 80° arc in a vertical plane? | | |
| | 3.3.1.4.1 | Is boom designed and constructed so as not to have any tail swing? | | |
| | 3.3.1.4.1 | Are means provided to prevent wedging or binding of the unloaded block or hook when at boom end? | | |
| | 3.3.1.4.1 | Is there no less than 3 steps to position boom? | | |
| | 3.3.1.4.1 | Is the crane truck designed and constructed so that no part of the boom or related structure will pass over the operator's head? | | |
| | 3.3.1.4.1 | Is beam length at least 12 feet retracted and 19 feet extended? | | |
| Winch | 3.3.1.4.2 | Does the winch consist of an electric motor and a gear reducer which is directly connected to the cable storage drum? | | |
| | 3.3.1.4.2 | Do reduction gears operate in a constant oil bath? | | |
| | 3.3.1.4.2 | Are magnetic filler and drainage plugs provided? | | |
| Cable storage drum | 3.3.1.4.3 | Does cable storage drum have a minimum tread diameter of 18 times the cable diameter? | | |
| | 3.3.1.4.3 | Is the width of the drum such that no more than 3 layers of cable are needed on the drum with boom fully collapsed and hook at boom end? | | |
| | 3.3.1.4.3 | Is the cable storage drum provided with a ramp thread (if required) on each flange to prevent piling of cable at flanges? | | |
| | 3.3.1.4.3 | Are flanges of sufficient depth to prevent cable from riding off the drum? | | |
| | 3.3.1.4.3 | Is drum designed to permit proper spooling of cable? | | |
| Sheaves | 3.3.1.4.4 | Are cable sheaves provided to guide the load line at each point of cable direction change? | | |
| | 3.3.1.4.4 | Is the minimum diameter of sheaves at least 18 times the cable diameter? | | |
| | 3.3.1.4.4 | Are sheaves mounted on bearings which are either grease sealed or have external lubrication points? | | |
| Cable | 3.3.1.4.5 | Is the cable for the load line 6 by 37 flexible hoisting rope? | | |
| | 3.3.1.4.5 | Is minimum factor of safety 5 times the tensile stresses induced by handling rated loads? (A certificate of compliance is required.) | | |
| | 3.3.1.4.5 | Is cable length sufficient to permit lowering hook to a position 4 feet below ground level when the boom is fully extended and raised, with not less than 3 dead turns on the cable storage drum? | | |
| | 3.3.1.4.5 | Is the cable attachment end fitted with a galvanized rope thimble or other suitable connection? | | |


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| *Component* | *Applicable Specification Paragraph* | *Item To Be Checked* | *Yes* | *No* |
|---|---|---|---|---|
| Block | 3.3.1.4.6 | Do the block and/or hook have adequate weight to insure payout of the cable without load on the hook when the hook is in any position? | | |
| | 3.3.1.4.6 | Is the hook the swivel type and is it fitted with a safety latch? | | |
| Limit switches | 3.3.1.6 | Are the vertical and horizontal movement of the boom and vertical travel of the load hook controlled by means of limit switches? | | |
| | 3.3.1.6 | Are mechanical stops provided at extreme limits of travel for boom hoist and sluing mechanisms? | | |
| Wheels | 3.3.2 | Are wheels flat faced with steel banded pressed on tires? | | |
| | 3.3.2 | Are there 2 single front wheels, and 2 single rear drive wheels? | | |
| | 3.3.2 | Do wheels, when equipped with new tires, clear any part of the truck structure under all conditions of operation as tested herein? | | |
| Tires | 3.3.2.1 | Are tires of the industrial, solid rubber high profile cushion type? | | |
| | 3.3.2.1 | Do tires have either smooth or non-directional type treads? | | |
| Steering | 3.3.3 | Is steering accomplished by an automotive type steering wheel, not exceeding 20 inches in diameter, mounted horizontally on the steering column and equipped with a spinner knob, which turns the front and rear wheels? | | |
| | 3.3.3 | Does clockwise or right hand rotation of the steering wheel provide for right hand or clockwise turning of the truck while going forward? | | |
| | 3.3.3 | Is provision made for precision adjustment to compensate for wear of all major steering parts, particularly steering gear reduction mechanism? | | |
| | 3.3.3 | Can friction points on steering mechanism be lubricated without disassembly or removal of any other component? | | |
| | 3.3.3 | Is a hydraulic steering booster or other power steering device provided? | | |
| Axles | 3.3.3.1 | Are means provided to prevent wheels or tires from contacting any part of the truck structure irrespective of obstacle magnitude? | | |
| | 3.3.3.1 | Are material requirements of this paragraph met? | | |
| Service brakes | 3.3.4.1 | Are service brakes of the hydraulically actuated internal expanding type or disc type mounted on the drive wheels? | | |
| | 3.3.4.1 | Is brake system interlocked with a relay switch which disconnects the electric circuit from the traction motor when brakes are applied? | | |
| | 3.3.4.1 | Are brakes applied by releasing brake pedal? | | |
| | 3.3.4.1 | Are brakes capable of retaining truck with maximum rated load on a 10 percent grade? | | |



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| *Component* | *Applicable Specification Paragraph* | *Item To Be Checked* | *Yes* | *No* |
|---|---|---|---|---|
| Service brakes (continued) | 3.3.4.1 | Is hydraulic master cylinder accessible for checking and refilling? | | |
| | 3.3.4.1 | Are service brakes protected to insure satisfactory operation in accordance with standard automotive practice? | | |
| | 3.3.4.1 | Can hydraulic brake system be bled of air without removing the wheels or disconnecting hydraulic lines? | | |
| | 3.3.4.1 | Is brake lining, when used, of molded type, riveted or cement bonded to the brake shoe? | | |
| Boom hoist, boom sluing and load line hoist brakes | 3.3.4.2 | Are boom hoist, boom sluing, and load line hoist brakes of the external contracting or internal expanding double shoe type or of the disc type? | | |
| | 3.3.4.2 | Are brakes mounted on the motors or drive shafts of the respective units? | | |
| | 3.3.4.2 | Is brake lining of molded type, riveted or cement bonded to the brake shoes or discs? | | |
| | 3.3.4.2 | Are boom hoist, boom sluing, and load line hoist brake systems interlocked with a relay switch which will apply the brakes, by means of a solenoid plunger, when the electrical circuit is disconnected? | | |
| | 3.3.4.2 | Is the load line hoist motor used to partially brake the speed of the load hook before application of friction brakes? | | |
| Operator's platform | 3.3.5.1 | Is operator's platform made from non-skid steel plate? | | |
| | 3.3.5.1 | Is operator's platform to the right of steering wheel? | | |
| | 3.3.5.1 | Does operator's platform have sufficient depth to permit operator to stand within the plan outline of the truck? | | |
| | 3.3.5.1 | Is platform enclosed on sides and rear with a plate extending at least 6 inches above the platform? | | |
| Fenders | 3.3.5.2 | If wheels protrude beyond the body of the truck, are suitable fenders provided? | | |
| Towing hooks | 3.3.5.3 | Are 2 ring or pin type towing hooks provided, 1 in the front and 1 in the rear of the crane truck? | | |
| | 3.3.5.3 | Are towing hooks recessed in the center front and the center rear of the truck so that they do not project from the truck? | | |
| Outriggers | 3.3.5.4 | Is crane truck equipped with 1 manually adjustable outrigger on each side? | | |
| | 3.3.5.4 | Are outriggers equipped with adjustable jacks? | | |
| | 3.3.5.4 | Is provision made for securing the jacks when not in use? | | |
| Hand holds | 3.3.5.6 | Are suitable hand holds and steps provided to assist operator in mounting the crane truck? | | |
| Controls | 3.4.1 | Are all switch and lever type controls except directional controls, provided with position markings at the switch or lever? | | |



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| *Component* | *Applicable Specification Paragraph* | *Item To Be Checked* | *Yes* | *No* |
|---|---|---|---|---|
| Controls (continued) | 3.4.1 | Are all controls and instrumentation protected against moisture and adverse weather conditions? | | |
| | 3.4.1 | Are directional controls provided with position markings either at the control lever or on a diagram visible to the operator? | | |
| | 3.4.1 | Are lever controls equipped with hand grip knobs? | | |
| | 3.4.1 | Is there at least ¾ of an inch clearance between the operator's hand and any adjacent knob, lever, fitting, or truck structure, when control knobs or levers are in any operating position? | | |
| Light switch | 3.4.1.1 | Is there an "on–off" headlight switch? | | |
| | 3.4.1.1 | Does it control the tail lamp, and the headlight, when required? | | |
| | 3.4.1.1 | Is there an "on–off" spotlight switch? | | |
| | 3.4.1.1 | Are switch locations convenient to the operator? | | |
| Brake pedal | 3.4.1.2 | Is brake pedal conveniently located for left foot operation? | | |
| Accelerator | 3.4.1.3 | Is accelerator control to the right of the operator and arranged for comfortable right hand operation? | | |
| Directional controls | 3.4.1.4 | Is a lever(s) for forward and rearward control located for right hand operation? | | |
| Crane controls | 3.4.1.5 | Are all crane controls self-centering levers? | | |
| | 3.4.1.5 | Are boom hoisting, load line hoisting, and boom sluing controls conveniently located for right hand operation? | | |
| | 3.4.1.5 | Does forward or downward motion of control levers lower load, lower boom, and swing boom clockwise? | | |
| | 3.4.1.5 | Is location of crane control levers, from left to right, as follows: load line hoist control lever, boom hoist control lever, and sluing control lever? | | |
| | 3.4.1.5 | Are controls designed to permit independent or simultaneous hoisting boom topping and boom sluing? | | |
| Hour meter | 3.4.2.1 | Is hour meter flush mounted in a location convenient to and plainly visible to the operator? | | |
| | 3.4.2.1 | Does hour meter register total number of operating hours? | | |
| | 3.4.2.1 | Does totalizer of hour meter register not less than 9999? | | |
| Headlight | 3.5.1 | Are headlights, automotive sealed beam type, of not less than 32 candlepower? | | |
| | 3.5.1 | Is directional focus of headlight, when provided adjustable both vertically and horizontally? | | |
| | 3.5.1 | Is headlight, when provided, protected against damage by position or guard? | | |
| Spotlight | 3.5.1.1 | Is there a spotlight of not less than 32 candlepower which is interchangeable with headlights? | | |



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| *Component* | *Applicable Specification Paragraph* | *Item To Be Checked* | *Yes* | *No* |
|---|---|---|---|---|
| Spotlight (continued) | 3.5.1.1 | Is spotlight directionally controllable by the operator without leaving the operator's platform? | | |
| Stoplight | 3.5.2 | Is a red, reflector type, combination stoplight and tail lamp, of not less than 15 and 3 candlepower respectively, recessed or mounted on the rear of the truck? | | |
| | 3.5.2 | Is stoplight recessed or mounted on the rear within the plan outline of the truck? | | |
| | 3.5.2 | Does stoplight operate only upon actuation of the service brake pedal? | | |
| | 3.5.2 | Is stoplight inoperative when parking brake is applied? | | |
| Horn | 3.5.3 | Is crane truck equipped with an electric horn with push button conveniently located for right foot or right hand operation? | | |
| | 3.5.3 | Is the horn, horn button, and electrical wiring protected against moisture and adverse weather conditions? | | |
| Fire extinguisher | 3.5.4 | If specified, is a fire extinguisher provided which meets the requirements of this paragraph? | | |
| Load line hoist lift speed | 3.6.1.1 | Is the hook lifting speed for lifting maximum rated load not less than:<br>14 or more than 22 feet per minute for size I<br>8 or more than 14 feet per minute for size II? | | |
| Load line hoist lowering speed | 3.6.1.2 | Is the hook lowering speed with load not less than:<br>20 feet per minute for size I<br>10 feet per minute for size II? | | |
| Boom topping speed | 3.6.2 | Is the boom, with rated load, capable of going from a horizontal position to maximum elevation, or vice versa, in not less than 60 seconds and not more than 90 seconds? | | |
| Boom sluing speed | 3.6.3 | Is time required for swinging the boom through 270° arc with rated load not less than 30 seconds or more than 50 seconds? | | |
| Turning radius | 3.6.6 | Is crane truck capable of turning in 90° intersecting aisles whose width is:<br>90 inches for size I<br>103 inches for size II? | | |
| Steering wheel | 3.6.7 | Is steering wheel capable of being turned from extreme left or vice versa, in not more than 7 complete turns of the hand steering wheel? | | |
| Hook clearance | 3.6.8 | Is maximum hook height, measured from the pull point of the hook to the ground, at least 17 feet with the hook 5 feet from front bumper with beam in fully extended position, and 7 feet with the hook at 14½ feet from front bumper? | | |
| Under-clearance | 3.6.9 | Is under-clearance sufficient to permit operating crane truck from one horizontal plane to another, up or down a 10 percent slope? | | |

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| *Component* | *Applicable Specification Paragraph* | *Item To Be Checked* | *Yes* | *No* |
|---|---|---|---|---|
| Over-all height | 3.6.10 | Is the crane truck capable of passing under an obstruction the underside of which is 91 inches above ground level? | | |
| Traveling capacity | 3.6.11 | Is the crane truck capable of traveling forward or rearward, in a straight line, over a level surface, with its boom at any setting and carrying the appropriate rated load for that setting? | | |
| Bearings | 3.7.1 | Do bearings comply with the applicable requirements of this paragraph? | | |
| Lubrication | 3.7.2 | Are ready accessible means provided for applying lubrication to all parts and components? | | |
| Workmanship | 3.7.3 | Are rivet connections, weld joints, bolt connections, castings and finish in compliance with the applicable paragraph? | | |
| Nameplate | 3.7.6.2 | Is a nameplate provided which meets the requirements of this paragraph? | | |
| Load capacity chart | 3.7.6.3 | Is there a corrosion resistant metal plate provided which meets the requirements of this paragraph? | | |
| Instruction plate | 3.7.6.4 | Is there a corrosion-resistant metal plate provided which meets the requirements of this paragraph? | | |
| Registration data | 3.7.7 | Is assigned registration number for each truck painted in black enamel letters 3 inches high on each side and in center of battery compartment? | | |
| Maintenance service, repair, and parts manuals | 3.7.9.1 | Is a copy of manufacturer's standard commercial manual included with truck or a more comprehensive manual as specified. | | |



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# PRE-TEST INSPECTION TEST CHECK SHEET
# CRANE TRUCK, WAREHOUSE, ELECTRIC, SOLID RUBBER TIRES

Description of Truck_____

Truck Mfgr._____ Date_____

Model_____ Test Location_____

Serial No._____ Recorded by_____

| *Component* | *Applicable Test Form* | *Item To Be Checked* | *Yes* | *No* |
|---|---|---|---|---|
| Steering wheel | 2G | Is the maximum gage reading required to turn the steering wheel in a counterclockwise direction 20 pounds or less? | | |
| | 2G | Is the maximum gage reading required to turn the steering wheel in a clockwise direction 20 pounds or less? | | |
| | 2G | Is there any evidence of permanent deformation to the steering wheel, column, related mechanical and/or hydraulic linkage? | | |
| Towing hooks | 5G | Can test truck be pulled, with wheels locked, without front towing hook failing? | | |
| | 5G | Can test truck be pulled, with wheels locked, without rear towing hook failing? | | |
| | 5G | Can test truck be pulled, with wheels locked, without visual observation of permanent distortion of front towing hook? | | |
| | 5G | Can test truck be pulled, with wheels locked, without visual observation of permanent distortion of rear towing hook? | | |
| Speed | 7G | Is average forward and rearward speed with maximum rated load at least 3½ miles per hour? | | |
| | 7G | Is average forward and rearward speed without load at least 4 miles per hour? | | |
| Slope ascension | 8G | Can truck, when carrying maximum rated load, stop on a 10 percent grade and start up again? | | |
| Stability | 14G | Were all phases of the stability test completed without failure, such as crane truck tipping? | | |
| | 14G | Were all phases of the stability test completed without visual observation of permanent distortion to any part of the crane truck? | | |
| Wear of steering mechanism | 15G | Is the free play of the steering wheel before the performance test not more than 5 inches? | | |
| Acceleration | 16G | Is average acceleration for first 22 feet from a standing start at least:<br>2.2 feet per second per second for truck without load?<br>2.0 feet per second per second for truck with maximum rated load? | | |
| Stopping distance | 17G | Is average stopping distance from rated speed not greater than 5 feet when normal braking is applied? | | |



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## TEST FORM NO. 2G
## STEERING WHEEL

### Crane Truck, Warehouse, Electric, Solid Rubber Tires

*Purpose:* To determine whether the force required to turn the steering wheel meets specifications.

*Test Description:* Place the unloaded truck on a level concrete surface. Using a tension dynamometer (with a 0-250 pound dial) or equivalent, equipped with a maximum hand, measure the force required to move the steering wheel.

*Test Procedure:*

1. Place steer wheels in a straight-ahead position.
2. Assemble an adapter pulley having a root diameter equal to the steering wheel diameter and means for attachment to the wheel spokes, to the steering wheel.
3. Attach a tension dynamometer to the adapter pulley.
4. Place truck on dry, level, smooth, trowelled or brushed concrete.
5. Grasp the dynamometer lead and apply a steady force, in a counterclockwise direction and in the wheel plane, until the wheel turns.
6. Observe the maximum gage reading within the first half of the inner steer wheel angle.
7. Observe the maximum gage reading at extreme ends of the steer.
8. Perform 5, 6, and 7 in a clockwise direction.

*Addendum Test*

After completing test requirements above conduct following:

1. Block steering wheels in straight-ahead position.
2. Grasp steering wheel and exert at least 200 pounds pressure in clockwise direction.
3. Repeat in counterclockwise direction.
4. Perform operations 2 and 3 four times.
5. Complete data form for addendum questions.

*Data Form:*

Description of Truck_____

Truck Mfgr._____ Date_____

Model_____ Test Location_____

Serial No._____ Recorded by_____

| *Question* | *Counter Clockwise Direction* | *Clockwise Direction* |
|---|---|---|
| Maximum gage first half of inner steer angle | | |
| Maximum gage reading at extreme ends of steer | | |

| *Addendum Question* | *Yes* | *No* |
|---|---|---|
| Is there any evidence of permanent deformation to steering wheel, related steering mechanism and/or hydraulic linkage | | |



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## TEST FORM NO. 5G
## TOWING HOOKS

### Crane Truck, Warehouse, Electric, Solid Rubber Tires

*Purpose:* To determine whether towing hooks have sufficient strength.

*Test Description:* Using steel cable or chain, attach to the front towing hook of the test truck another truck of the same or greater capacity. Lock wheels on test truck and pull with the other truck on dry, level concrete until test truck slides forward not more than 1 foot. Repeat test on rear towing hook.

*Test Procedure:*

1. Place test truck on dry, level, smooth, trowelled or brushed concrete.
2. Use steel cable or chain to attach a second truck of same or greater capacity to front towing hook of test truck.
3. Release brake pedal of test truck to lock wheels and starting second truck slowly, pull on towing hook until test truck slides forward not more than 1 foot, with wheels locked.
4. Perform operations 2 and 3, testing rear towing hook.
5. Complete data form.

*Data Form:*

Description of Truck_____

Truck Mfgr._____ Date_____

Model_____ Test Location_____

Serial No._____ Recorded by_____

| Question | Yes | No |
|---|---|---|
| Can test truck be pulled, with wheels locked, without front towing hook failing? | | |
| Can test truck be pulled, with wheels locked, without rear towing hook failing? | | |
| Can test truck be pulled, with wheels locked, without visual observation of permanent distortion of front towing hook? | | |
| Can test truck be pulled, with wheels locked, without visual observation of permanent distortion of rear towing hook? | | |
| Is there any evidence of permanent distortion or deformation of the frame? | | |



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## TEST FORM NO. 7G
## SPEED

### Crane Truck, Warehouse, Electric, Solid Rubber Tires

*Purpose:* To determine whether truck on a level surface is capable of a forward or rearward speed of not less than 3½ miles per hour with rated load and 4 miles per hour without load.

*Test Description:* Test is conducted on a 188-foot or more paved level course. Truck, with rated load, is accelerated through a 50-foot or more approached, timed through an 88-foot test length, and then brought to a stop. (This 88-foot length is to simplify calculations as 88 feet-minute is 1 m.p.h.) Three runs are made forward in high speed and 3 in reverse. The data is recorded and average speeds determined. The test is repeated with truck unloaded.

*Course Layout:*

A ⟵ 50′ or more ⟶ B ⟵ 88′ ⟶ C ⟵ 50′ or more ⟶ D

*Test Procedure:*

1. Install 1 or 2 fully charged batteries, as required, of the size used by the Government for the size truck under test and then operate truck for 1 hour prior to test.
2. Locate truck, carrying maximum rated load, for forward run with front wheels on line A. Load shall be attached to the truck by 2 guy lines leading from the load to opposite sides of the truck when they are anchored to the truck structure.
3. Start forward test run and have truck at speed by the time it reaches line B.
4. Using split second stop watch, determine time in seconds used in traveling from line B to line C and record on data form.
5. Perform operations 3 and 4 a second and third time.
6. Locate truck, carrying rated load, for rearward run with rear wheels on line A.
7. Start rearward test run and have truck at speed by the time it reaches line B.
8. Using split second stop watch, determine time in seconds used in traveling from line B to line C and record on data form.
9. Perform operations 7 and 8 a second and third time.
10. Complete data form.



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*Data Form:*

Description of Truck _____

Truck Mfgr. _____ Date _____

Model _____ Test Location _____

Serial No. _____ Recorded by _____

| | Forward | | Rearward | |
|---|---|---|---|---|
| | with load | without load | with load | without load |
| Time in seconds, No. 1 test | | | | |
| Time in seconds, No. 2 test | | | | |
| Time in seconds, No. 3 test | | | | |
| Total time | | | | |
| Average time (⅓ of total time) | | | | |
| Average speed in m.p.h. $\frac{60}{\text{average time}}$ | | | | |

| Question | Yes | No |
|---|---|---|
| Is average forward and rearward speed with maximum rated load at least 3½ m.p.h.? | | |
| Is average forward and rearward speed without load at least 4 m.p.h.? | | |



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## TEST FORM NO. 8G
## SLOPE ASCENSION

### Crane Truck, Warehouse, Electric, Solid Rubber Tires

*Purpose:* To determine whether crane truck, when carrying maximum rated load, can accelerate from a dead stop on a 10 percent grade.

*Test Description:* Crane truck, carrying maximum rated load, ascends a 10 percent grade in a forward direction on a dry concrete surface. Crane truck stops and starts up slope again.

*Test Procedure:*

1. Place maximum rated load on load hook.
2. Drive crane truck up 10 percent ramp in a forward direction and bring it to a dead stop.
3. Start crane truck from the dead stop and proceed up ramp.
4. Complete data form.

*Data Form:*

Description of Truck_____

Truck Mfgr._____ Date_____

Model_____ Test Location_____

Serial No._____ Recorded by_____

| Question | Yes | No |
|---|---|---|
| Can crane truck, when carrying maximum rated load, stop on 10 percent grade and start up again? | | |



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## TEST FORM NO. 14G
## STABILITY

### Crane Truck, Warehouse, Electric, Solid Rubber Tires

*Purpose:* To determine whether crane truck has sufficient stability and whether it meets hoisting, topping, and sluing specifications.

*Test Description:* Test standard load for crane truck type is placed on ground 60 inches directly in front of the front bumper. Load is picked up by hoist and, without changing boom topping, is slued 135° to the right. Load is lowered to ground, and then picked up and raised until 2 feet above the ground. Boom is elevated to raise load another foot and lowered to return load to the 2-foot level. The boom is then turned 15° towards the left and the test repeated. This procedure is followed at 15° increments until boom is slued 135° to the left.

*Test Procedure:*

1. Place maximum rated load for the type crane truck being tested directly in front of crane truck.
2. Attach hook of crane truck to eye of test load and slowly raise load vertically until bottom is 2 feet above ground level.
3. Without changing boom topping, slowly slue boom 135° to right.
4. Without changing boom topping, lower load to ground level.
5. Without changing boom topping, raise load until bottom of load is 2 feet above ground level.
6. Elevate boom until bottom of load is 3 feet above ground.
7. Lower boom until bottom of load is returned to 2-foot elevation.
8. Without changing boom topping, slue boom 15° towards left.
9. Repeat operations 4, 5, 6, 7, and 8 at 15° increments until boom is slued 135° to the left.
10. Complete data form.

*Data Form:*

Description of Truck_____

Truck Mfgr_____ Date_____

Model_____ Test Location_____

Serial No._____ Recorded by_____

| Question | Yes | No |
|---|---|---|
| Were all phases of the stability test completed without failure, such as truck tipping, load falling, etc.? | | |
| Were all phases of the stability test completed without visual observation of permanent distortion to any part of the crane truck? | | |



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## TEST FORM NO. 15G
## WEAR OF STEERING MECHANISM

### Crane Truck, Warehouse, Electric, Solid Rubber Tires

*Purpose:* To determine cumulative wear in the steering mechanism of the truck due to carrying out requirements of the performance test.

*Test Description:* With the truck stationary and with no load, the amount of free play in the steering wheel is measured. This procedure is followed before and after the performance test. If an abnormal increase in free play is noted after the performance test is run, it is an indication of excessive wear in the steering mechanism.

*Test Procedure:*

1. Place unloaded truck on a dry, level, concrete surface with its rear wheels parallel to the longitudinal axis of the truck.
2. Attach to the steering column, immediately below the steering wheel and parallel to it, a piece of stiff fiber board or similar material which is semi-circular in shape and which has a radius at least 1 inch greater than the radius of the steering wheel.
3. Mark a reference point on the upper portion of the outer edge of the steering wheel; mark a corresponding point on the fiber board templet.
4. Turn the steering wheel counterclockwise until the wheels of the truck begin to turn. Mark a point on the fiber templet corresponding to the reference point on the outer edge of the steering wheel.
5. Turn the steering wheel clockwise until the wheels of the truck begin to turn. Mark a point on the fiber board templet corresponding to the reference point on the outer edge of the steering wheel.
6. Remove the fiber board templet from the truck.
7. Place the fiber board templet on a flat surface. Measure the distance between the two points made in accordance with operation numbers 4 and 5 above. This distance, in inches, is the free play of the steering wheel.
8. Complete data form.



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*Data Form:*

Description of Truck_____

Truck Mfgr._____ Date_____

Model_____ Test Location_____

Serial No._____ Recorded by_____

Free Play of Steering Wheel:

Before Performance Test_____inches

After Performance Test_____inches

| Question | Yes | No |
|---|---|---|
| Is the free play of the steering wheel before the performance test not more than 5 inches? | | |
| Is the free play of the steering wheel after the performance test not more than 1.5 times the free play of the steering wheel before the performance test? | | |



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## TEST FORM NO. 16G
## ACCELERATION

### Crane Truck, Warehouse, Electric, Solid Rubber Tires

*Purpose:* To determine acceleration of truck, with and without load, on a level surface from a standing start.

*Test Description:* Test is conducted on an 88 foot paved level course. Truck is accelerated at maximum rate from a standing start, timed through a distance of 22 feet and then brought to a stop. (This distance is used so that this test can be performed in the same area as used for the Speed Test.)

*Course Layout:*

A ■ ····· 22′ ·····■ B ■····· 66″ ·····■ C

*Test Procedure:*

1. Operate truck for 60 minutes prior to test.
2. Locate truck, without load, with front wheels on line A.
3. Station an observer at line B. The observer shall be provided with a split second stop watch.
4. The test leader shall be stationed at line A.
5. Test leader shall start test by calling "mark". When this signal is given the truck operator starts the truck and the observer starts his watch.
6. The truck operator will maintain maximum truck acceleration through the entire 22 foot test run.
7. As the front wheels of the truck pass over line B, the observer stops his watch.
8. Record stop watch times on data form.
9. Perform operations 1 through 8 a second and third time.
10. Perform operations 1 through 9 with truck carrying maximum load. Attach load to truck by 2 guy lines leading from the load to opposite sides of the truck where they are anchored to the truck structure.
11. Complete data form.

*Data Form:*

Description of Truck_____

Truck Mfgr._____ Date_____

Model_____ Test Location_____

Serial No._____ Recorded by_____

| Question | Yes | No |
|---|---|---|
| Is average acceleration for the first 22 feet at least:<br>2.2 feet per second per second for truck without load?<br>2.0 feet per second per second for truck with maximum rated load? | | |



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## TEST FORM NO. 17G
## STOPPING DISTANCE

### Crane Truck, Warehouse, Electric, Solid Rubber Tires

*Purpose:* To determine truck can be stopped within a given distance from specified maximum speeds.

*Test Description:* Run truck at maximum speed over dry, smooth, trowelled or brushed concrete, apply brakes and measure the distance required to stop the truck.

*Course Layout:*

A ■ ····· 100′ or more ····· ■ B ■ ····· 88′ ····· ■ C

*Test Procedure:*

1. Place a clearly visible vertical mark on the side of the truck directly beneath the normal position of the operator's head. Mark shall be such that the operator can see it easily.
2. Operate truck for 15 minutes prior to test. Operate brakes at least 10 times.
3. Run the loaded truck in forward direction onto a strip of dry, smooth, level, trowelled or brushed concrete, measured and marked as specified above in "Course Layout."
4. Start 2 split second stop watches at same time so that they are synchronized.
5. Station an observer at line B and a second observer at line C. Each observer shall be provided with one of the stop watches described in operation 4.
6. Start the truck at line A and accelerate it so that it has attained full speed before it reaches line B.
7. Maintain the truck at full speed between lines B and C.
8. As the front wheels of the truck pass over line B, the first observer stops his watch.
9. As the front wheels of the truck pass over line C, the second observer stops his watch.
10. The operator applies truck brakes, when the line placed on the truck in operation 1 is directly over line C.
11. After truck is stopped, measure the distance from line C to the line on the truck and record on data form.
12. Record watch readings on data form.
13. Perform operations 3 through 12 a second and third time.
14. Complete data form.



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*Data Form:*

Description of Truck _____

Truck Mfgr. _____ Date _____

Model _____ Test Location _____

Serial No. _____ Recorded by _____

| | *Watch Reading Line B* | *Watch Reading Line C* | *Difference in Watch Readings Line B— Line C* | *Stopping Distance in Feet* |
|---|---|---|---|---|
| Test No. 1 | | | | |
| Test No. 2 | | | | |
| Test No. 3 | | | | |
| Total Readings | | | | |
| Average Readings (⅓ of total readings) | | | | |
| Average Miles Per Hour (60 + difference in watch readings) | | | | |

| *Question* | *Yes* | *No* |
|---|---|---|
| Is average forward speed with load as specified in Test Form No. 7G? | | |
| Is average stopping distance with load *not* greater than 5 feet, when speed of truck is 3½ MPH? | | |



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5.2 **Performance test.**

5.2.1 *Layout of course.*

5.2.2.1 *Description.* The test course shall be set up in accordance with the provisions of Drawing E–4. Aisle width to be used shall be 9 feet. The entire perimeter of the test course shall be encompassed with suitable materials to form a barrier having a height of at least 12 inches.

5.2.1.2 *Course illustrated by Drawing E–4.*

5.2.1.3 *Specification of course features.*

5.2.1.3.1 *Obstacle test.* The obstacle test shall be set up in accordance with the provisions of Drawing F, as a part of the test course. Iron or steel blocks may be used. A guide line 15 feet long shall be provided to the left of the center line of the course and parallel to it. The inner edge of this guide line shall be at a distance from the course center line equal to ½ of the truck width, plus 6 inches. This spacing and arrangement will allow the left wheels of the machine under test to pass over the first block and then the right wheels to pass over the second block while the machine is moving forward in a straight line parallel to the guide line.

5.2.1.3.2 *Ramp design.* The ramp shall be set up in accordance with the provisions of Drawing G, as a part of the test course.

5.2.1.3.3 *Specification of standard test load.* Test loads used in the performance test shall be in accordance with 5.1.2.1 and Drawing L, M, and N.

5.2.1.3.4 *Specification of load areas.* Test standard loads in Positions 1, 2, and 3 shall be located in accordance with Drawing P. They shall be placed in frames made of 6 by 6 inch timbers, which have an inside measurement of 52 by 52 inches. Frames shall be attached to the floor in the positions shown on Drawings E–4 and P.

Test standard loads in Positions 4, 5, and 6 shall be located in accordance with Drawing Q. They shall be placed in frames made of 6 by 6 inch timbers, which have an inside measurement of 52 by 52 inches. Frames shall be attached to the floor in the positions shown on Drawings E–4 and Q.

5.2.1.3.5 *Lighting requirements.* Lights of the machine under test shall be on at all times during the test.

5.2.1.3.5.1 The headlights shall be turned off and turned on at the commencement of each lap and at each loading area.

5.2.1.3.6 *Course surface.* Except where otherwise specified the course shall be paved with concrete, or other material having a similar coefficient of friction.

5.2.2 *Standard procedures.*

5.2.2.1 *Changing batteries.* Each machine undergoing test shall be allowed not more than 2 battery changes during the course of each operating 8-hour day of the performance test. However, the time required to change batteries under the provisions of this paragraph shall be excluded from operating time. Batteries to be used by the machine under test shall be those specified or in use by the Department of Defense for the type and size of the machine under test.

5.2.2.2 *Speed of negotiating course.*

5.2.2.2.1 *General.* Each machine undergoing test should traverse each part of the performance test course at the maximum safe practical speed.

5.2.2.2.2 *Laps per hour.* During each individual operating hour of the performance test, each machine under test shall complete 6 circuits of the test course.

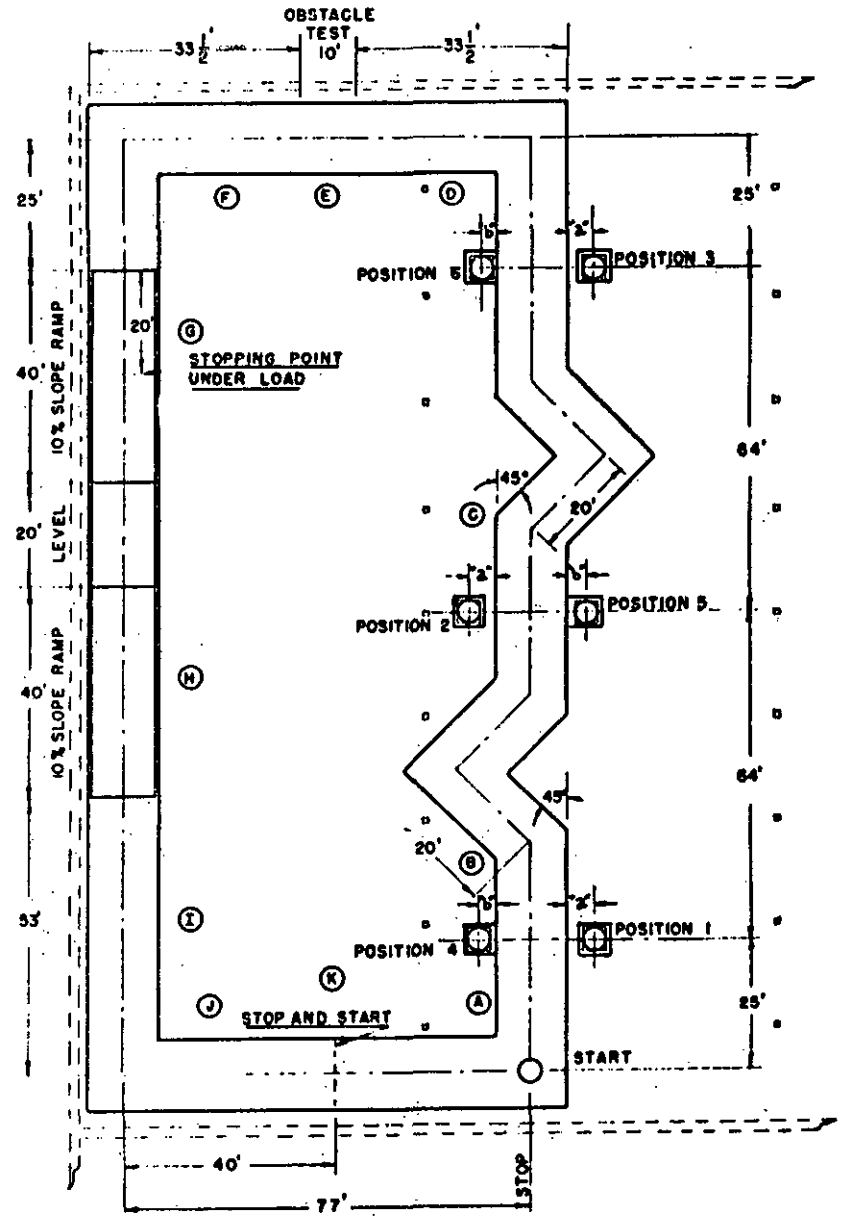
5.2.2.2.3 *Laps per 8-hour day.* During each individual operating 8-hour day of the performance test, each machine under test shall complete 48 circuits of the test course.



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DRAWING "E-4"

PROPOSED INDOOR TEST COURSE FOR CRANE TRUCKS



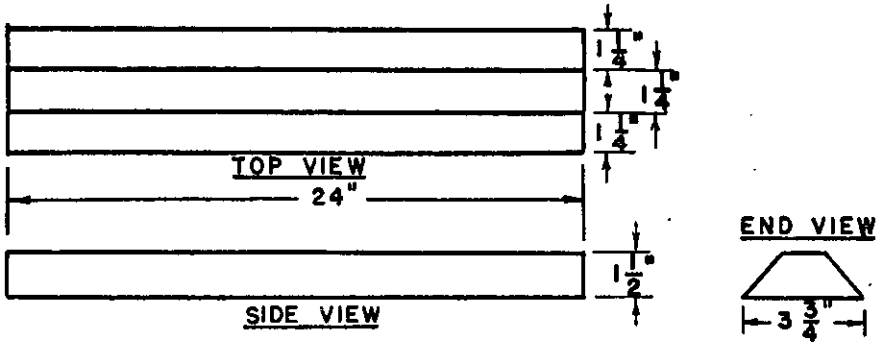

NOTES: "a" = 60 INCHES FROM FRONT BUMPER WHEN CRANE IS SPOTTED AT AISLE EDGE NEAREST LOAD
"b" = 36 INCHES FROM FRONT BUMPER WHEN CRANE IS SPOTTED AT AISLE EDGE NEAREST LOAD
LIGHTING TO BE A MINIMUM OF 5 FOOT CANDLES AT OPERATING LEVEL
TEST COURSE IS FINE-FLOAT FINISH CONCRETE OR EQUIVALENT
TEST COURSE IS SHOWN LAID OUT LENGTHWISE IN A STANDARD 200' BY 800' NAVY WAREHOUSE
Ⓓ, ETC. = SECTION DESIGNATION



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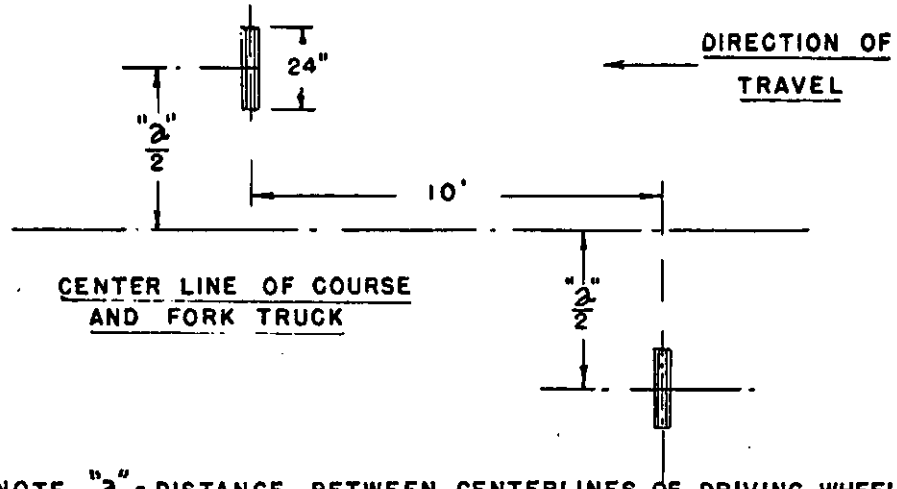
# DRAWING "F"

## DETAILS OF OBSTACLE BLOCK CONSTRUCTION




NOTE. 2 BLOCKS REQUIRED
BLOCKS TO BE ATTACHED TO TEST COURSE AS SHOWN BELOW

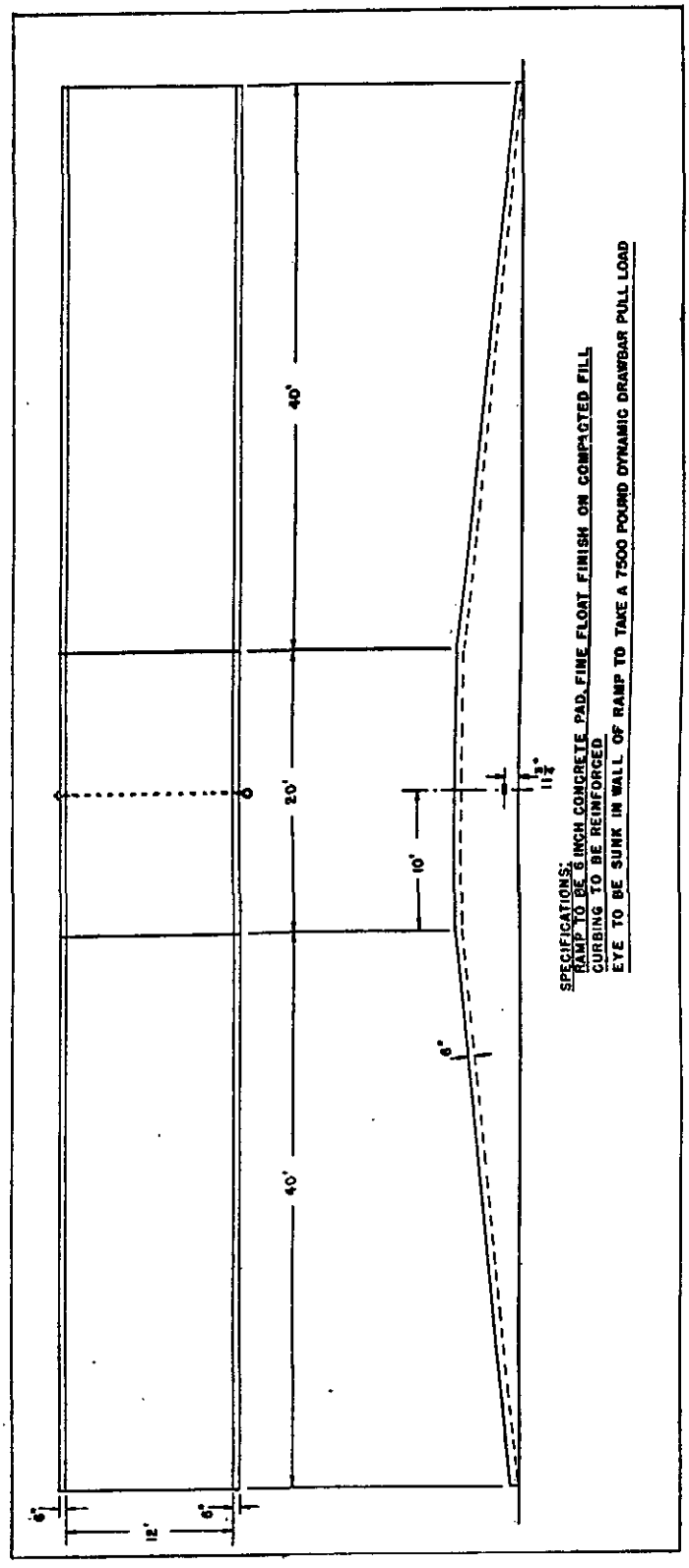
## LAYOUT OF OBSTACLE TEST AREA "E" OF TEST COURSE




NOTE. "a" = DISTANCE BETWEEN CENTERLINES OF DRIVING WHEELS



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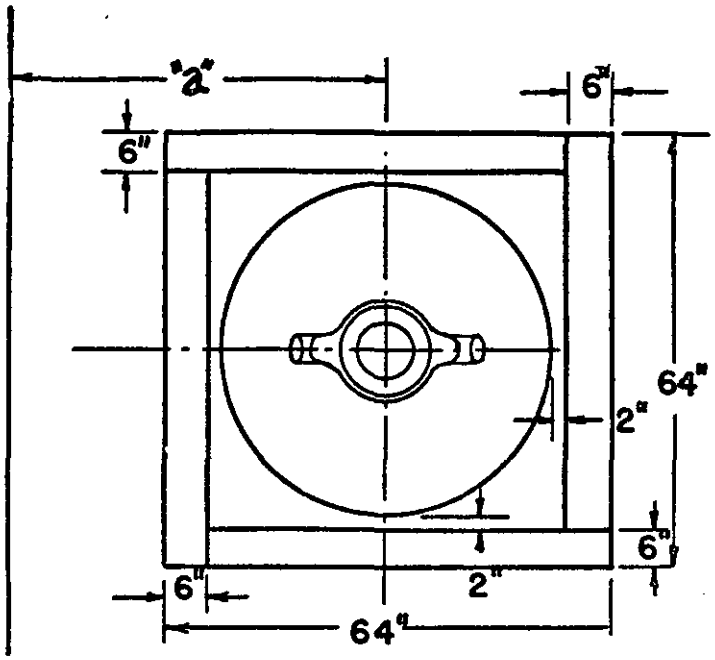



DRAWING "G" RAMP DESIGN



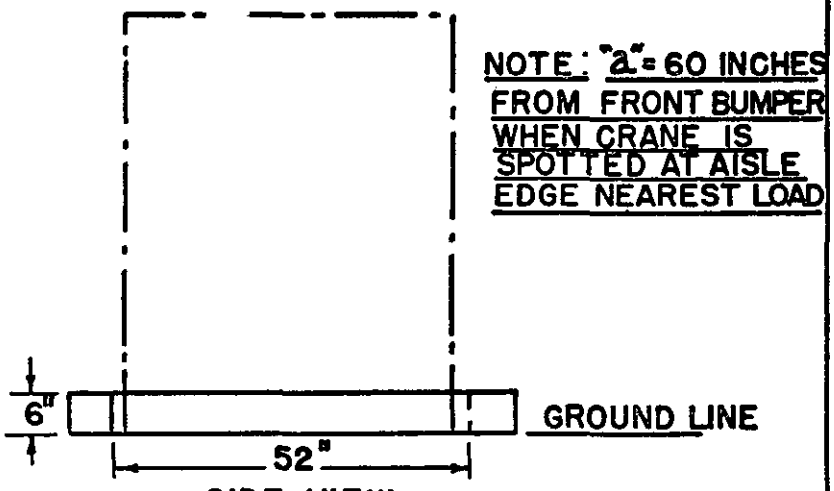
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## DETAILS OF LOAD POSITIONS 1, 2, AND 3
## DRAWING "E-4"




TOP VIEW

STANDARD TEST LOAD AS SHOWN IN DRAWINGS L, M, AND N




NOTE: "a" = 60 INCHES FROM FRONT BUMPER WHEN CRANE IS SPOTTED AT AISLE EDGE NEAREST LOAD

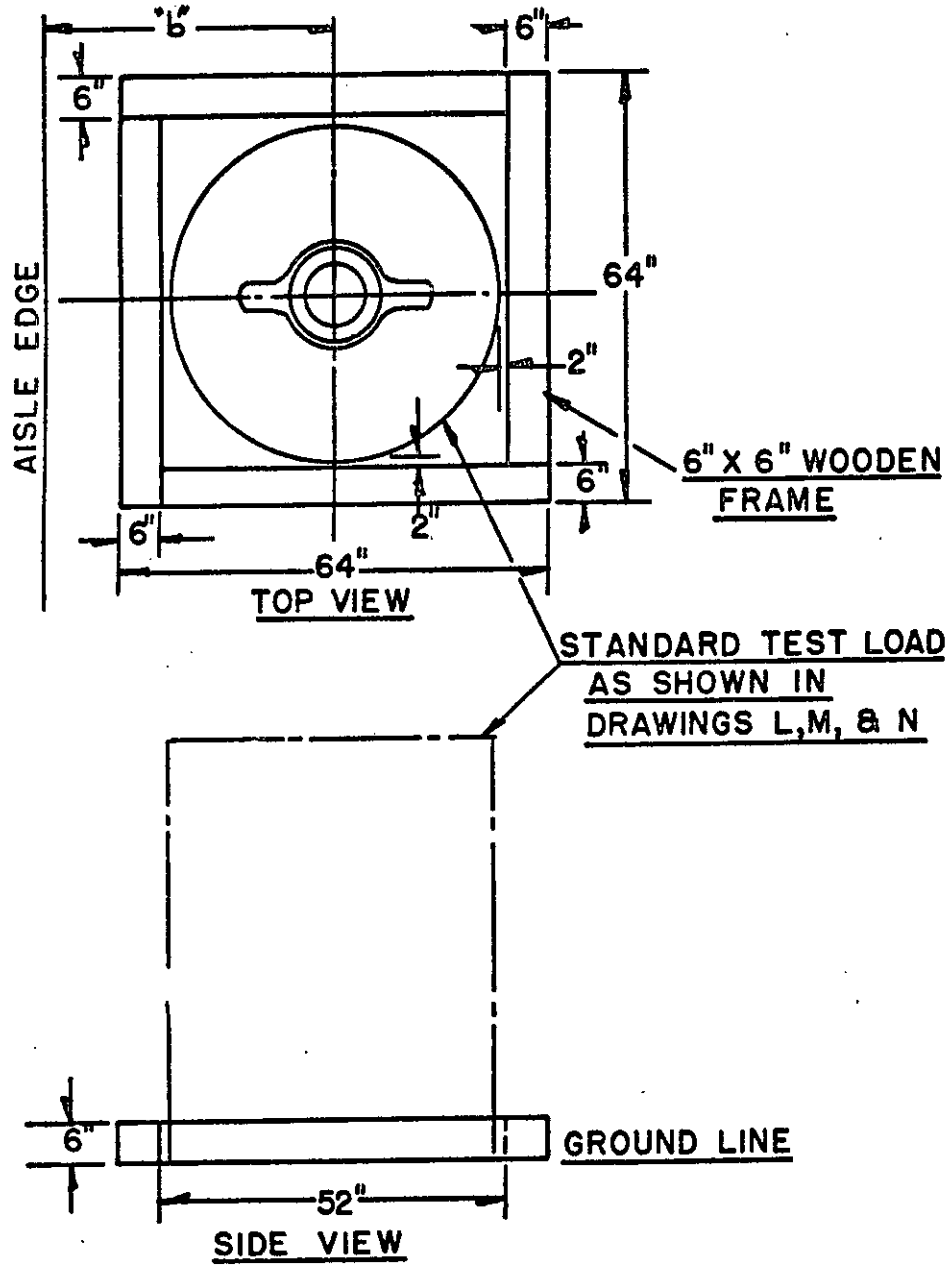
SIDE VIEW

DRAWING "P"



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DETAILS OF LOAD POSITIONS 4, 5, AND 6
DRAWING "E-4"




NOTE. "b"=36" FROM FRONT BUMPER WHEN CRANE IS SPOTTED AT AISLE EDGE NEAREST LOAD

DRAWING Q



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5.2.2.2.4 *Duration of the performance test.* The performance test specifid in this standard shall be conducted 8 hours per day, 5 days per week, for a period of 3 weeks, or a total of 15 operating 8-hour days of test. By means of the accelerated wear features built into the performance test, this period is intended to be the equivalent of approximately 3.5 months of normal cran truck operation.

5.2.2.3 *Conditions of course during test.* The test course shall be dry, clean, and free of any non-planned obstacles or foreign matrial during the course of the test.

5.2.2.4 *Periodic maintenance.* On the 4th, 8th, and 12th operating days of the performance test the "Preventive Maintenance Service and Inspection (Schdule "A"), shall be conducted according to the provisions of "Preventive Maintenance of Materials Handling Equipment" (NAVSANDA Publication 249).

This procedure shall be carried out after the machine under test has completed performance test requirements on the days specified. However, no repairs or replacement of parts shall be permitted.

5.2.2.5 *Method of negotiating course.* The machine under test shall begin each lap at the point labeled "START" on Drawing E–4. As the operator starts the machine under test, he shall operate the horn for approximately 1 second, turn off and turn on the headlights. In lap number 1 it shall proceed forward along the portion of the course marked "A" until it reaches the first load area. A test standard load shall be placed in this location. The machine under test shall slue 90° to the right and remove the test standard load from Position 1. It shall slue 90° to the left and proceed along the main aisle in the portion of the course marked "B" until it reaches Position 2.

The machine under test shall slue 90° to the left and set down the test standard load in Position 2. It then shall slue 90° to the right and proceed forward along the main aisle in the portion of the course marked "C" until it reaches Position 3.

The machine under test shall slue 90° to the right and pick up the test standard load in Position 3. It then shall slue 90° to the left and proceed forward along the main aisle in the portion of the course marked "D" until it reaches the obstacle test in the portion of the course marked "E".

The obstacle test shall be traversed by the machine under test being driven through this area in a straight line parallel to the guide line so that its left wheels pass over the first block and its right wheels pass over the second block. This part of the test may be conducted at low speed. However, the machine under test shall not be stopped prior to or while going over the blocks.

The machine under test shall proceed further along the main aisle in the portion of the course marked "F" and "G" until it reaches the position on the ramp labeled "Stopping Point" on the drawing. Here the driver shall stop the machine and then resume mounting the ramp. After the top of the ramp is reached he shall proceed along the main aisle in the portion of the course marked "H", "I", and "J".

When the machine under test reaches the portion of the course marked "K" on the drawing, the driver shall stop it and then restart it. After completion of this portion of the test he shall proceed along the main aisle to the point marked "Stop" on the drawing, where he shall again stop the machine. All stops shall be at the maximum safe deceleration rate.

In the second lap the machine under test shall re-enter the portion of the course marked "A" on the drawing. In portions "A", "B", "C", and "D" the procedure shall be the same as described for lap number 1, except that the machine under test shall set down a test load at each point where previously it was indicated that the machine picked one up, and pick up a test load where


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previously it was indicated that one should be set down. Maneuvers in the balance of the course, and the order in which the course is covered beyond the portion marked "D" on the drawing, shall be the same as described for lap number 1.

Lap number 3 shall be identical with lap number 1, except that Position 4 shall be substituted for Position 1, Position 5 shall be substituted for Position 2, and Position 6 shall be substituted for Position 3. Also, the indicated direction of sluing at each of the three positions in lap number 1 shall be reversed in lap number 3.

Lap number 4 shall be identical with lap number 2, except that references in the description of lap number 2 to lap number 1 should be changed to lap number 3.

The driver of the machine under test shall keep the boom in the direction the machine under test is travelling and within the permissible aisle limits at all times while traversing the performance course.

The "Hourly Time Record Sheet" on the following page shall be filled out for each operating hour of the performance test.

5.2.2.6 *Operating limitations.* The crane truck under test shall not be permitted to leave the bounds circumscribed in Drawing E–4, nor shall the procedure requirements specified herein be altered.

**5.3 Post-test.**

5.3.1 *When given.* The post test shall be conducted after a machine has completed the performance test or, if it fails the performance test, the inspection portion of this procedure shall be conducted after such a failure.

5.3.2 *Composition.* The post-test consists of 2 parts:

A. Post Test Inspection Test
B. Post Test Inspection Form

5.3.3 *Description of the Post Test Inspection Tests.* These tests shall be performed in accordance with the following schedule:

| *Applicable Test Form* | *Test Title* |
|---|---|
| 2G | Steering Wheel |
| 7G | Travel Speed |
| 15G | Wear of Steering Mechanism |
| 16G | Acceleration |
| 17G | Stopping Distance |

5.3.3.1 *How recorded.* Each of the questions contained in the Post-Test Inspection Test Check Sheet shall be answered by personnel designated by the Test Officer by placing an "X" in the column headed "Yes" or "No" as appropriate.

5.3.3.2 *How evaluated.* After completion of the Post-Test Inspection Test Check Sheet any "X" in the "No" column shall be cause for rejection of the truck.

5.3.4 *Post-Test Inspection Form.* Personnel under the direction of the Test Officer shall examine the truck under test, make applicable notation in the Post-Test Inspection Form, and shall perform overload tests required in Specification MIL–C–3988 utilizing strain gages and similar equipment to determine deformation.



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HOURLY TIME RECORD SHEET Date_____

Truck Manufacturer_____ Clock Readings:

Model_____ Test Location_____ Start_____

Serial No._____ Recorded by_____ Finish_____

| Lap Number | Watch Reading | Elapsed Time | Remarks |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |
| 6 | | | |
| 7 | | | |
| 8 | | | |
| 9 | | | |
| 10 | | | |
| 11 | | | |
| 12 | | | |
| 13 | | | |
| 14 | | | |
| 15 | | | |
| 16 | | | |
| 17 | | | |
| 18 | | | |
| 19 | | | |
| 20 | | | |
| Total | | | |

*Instructions for Timing Performance Test*

1. Start decimal minute stop watch at zero at beginning of each operating hour.
2. Record watch reading at end of each lap in "Watch Reading" space.
3. Subtract watch reading at end of preceding lap from reading at end of lap. Record difference in "Elapsed Time" space.
4. If truck under test completes 20 laps prior to completion of an operating hour, truck is to be stopped until start of next hour, and then resume performance test.

5.3.5 *Commutator damage.* During the course of the post-test inspection, maintenance personnel shall examine the commutator of the main drive motor(s). If any pitting is found in this part, the Test Officer shall recommend rejection of the machine under test. A report of the findings of this examination shall be prepared by the Test Officer and shall become a portion of the official test report of the machine under test.



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# POST-TEST INSPECTION TEST CHECK SHEET
# CRANE TRUCK, WAREHOUSE, ELECTRIC, SOLID RUBBER TIRES

Description of Truck_____

Truck Mfgr._____ Date_____

Model_____ Test Location_____

Serial No._____ Recorded by_____

| *Component* | *Applicable Test Form* | *Item To Be Checked* | *Yes* | *No* |
|---|---|---|---|---|
| Steering wheel | 2G | Is the maximum gage reading required to turn the steering wheel in a counterclockwise direction 20 pounds or less? | | |
| | 2G | Is the maximum gage reading required to turn the steering wheel in a clockwise direction 20 pounds or less? | | |
| | 2G | Is there any evidence of permanent deformation to steering wheel, column, related mechanical and/or hydraulic linkage. | | |
| Speed | 7G | Is average forward speed with maximum rated load at least 3½ miles per hour? | | |
| | 7G | Is average forward speed without load at least 4 miles per hour? | | |
| Wear of steering mechanism | 15G | Is the free play of the steering wheel after the performance test not more than 1.5 times the free play of the steering wheel before the performance test? | | |
| Acceleration | 16G | Is average acceleration for first 22 feet from a standing start at least:<br>2.2 feet per second per second for truck without load?<br>2.0 feet per second per second for truck with maximum rated load? | | |
| Stopping distance | 17G | Is average stopping distance from rated speed not greater than 5 feet when normal braking is applied? | | |



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## POST-TEST INSPECTION FORM

INSTRUCTIONS: Examine each item below in sufficient detail to determine any defects or excessive wear which have caused failure in test or might cause failure in normal operation. Place a check mark in column No. 1 or No. 2. If column No. 2 is checked, also check column Nos. 3, 4, or 5.

| Description | (1) *OK* | (2) Not *OK* | (3) *Adjust* | (4) *Repair* | (5) *Replace* |
|---|---|---|---|---|---|
| 1. Batteries | | | | | |
| 2. Brake | | | | | |
| 3. Clutch | | | | | |
| 4. Contacts | | | | | |
| 5 Cooling | | | | | |
| 6. Differential | | | | | |
| 7. Drive Axle | | | | | |
| 8. Drive Shaft | | | | | |
| 9. Electric Components | | | | | |
| a. Battery Blocking | | | | | |
| b. Contactor | | | | | |
| c. Controller | | | | | |
| d. Collector Ring | | | | | |
| e. Foot Pedal Control | | | | | |
| f. Interlocker | | | | | |
| g. Instruments | | | | | |
| h. Limit Switch | | | | | |
| i. Line Breaker | | | | | |
| j. Plug and Receptacle | | | | | |
| k. Power Unit (*Motors, etc.*) | | | | | |
| l. Resistor | | | | | |
| m. Seat and Brake Switch | | | | | |
| n. Speed Control | | | | | |
| o. Wiring | | | | | |
| 10. Engine | | | | | |
| 11. Forks | | | | | |
| 12. Frame and Body | | | | | |



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POST-TEST INSPECTION FORM (Continued)

| Description | (1) OK | (2) Not OK | (3) Adjust | (4) Repair | (5) Replace |
|---|---|---|---|---|---|
| 13. Fuel System | | | | | |
| 14. Hydraulic | | | | | |
| a. Control Levers and Valves | | | | | |
| b. Crosshead and Pull-Down Rod | | | | | |
| c. Lift Cylinder | | | | | |
| d. Lift Chains | | | | | |
| e. Lines and Fittings | | | | | |
| f. Mast Carriage and Rollers | | | | | |
| g. Mast Column | | | | | |
| h. Oil Tank | | | | | |
| i. Pump | | | | | |
| j. Tilt Cylinder | | | | | |
| 15. Muffler | | | | | |
| 16. Springs | | | | | |
| 17. Steering | | | | | |
| 18. Transmission | | | | | |
| 19. Universal | | | | | |
| 20. Wheels | | | | | |
| 21. Wheel Steer Parts (Electric) | | | | | |

NOTE: Place remarks opposite the proper number on the reverse side of this form. If more space is required, attach separate sheet. Remarks should include details of adjustments, repairs, or replacements indicated by check marks above.

| | |
|---|---|
| Truck Mfgr._____ | Date_____ |
| Model_____ | Test Location_____ |
| Serial No._____ | Mechanic_____ |
| Maximum lift height_____ | Payroll No._____ |



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POST-TEST INSPECTION FORM (Continued)

| | |
|---|---|
| 1. Batteries | |
| 2. Brake | |
| 3. Clutch | |
| 4. Contacts | |
| 5 Cooling | |
| 6. Differential | |
| 7. Drive Axle | |
| 8. Drive Shaft | |
| 9. Electric Components | |
| a. Battery Blocking | |
| b. Contactor | |
| c. Controller | |
| d. Collector Ring | |
| e. Foot Pedal Control | |
| f. Interlocker | |
| g. Instruments | |
| h. Limit Switch | |
| i. Line Breaker | |
| j. Plug and Receptacle | |
| k. Power Unit (*Motors, etc.*) | |
| l. Resistor | |
| m. Seat and Brake Switch | |
| n. Speed Control | |
| o. Wiring | |
| 10. Engine | |
| 11. Forks | |
| 12. Frame and Body | |
| 13. Fuel System | |
| 14. Hydraulic | |
| a. Control Levers and Valves | |
| b. Crosshead and Pull-Down Rod | |
| c. Lift Cylinder | |



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POST-TEST INSPECTION FORM (Continued)

| | |
|---|---|
| 14. *Hydraulic (continued)* | |
| d. Lift Chains | |
| e. Lines and Fittings | |
| f. Mast Carriage and Rollers | |
| g. Mast Column | |
| h. Oil Tank | |
| i. Pump | |
| j. Tilt Cylinder | |
| 15. Muffler | |
| 16. Springs | |
| 17. Steering | |
| 18. Transmission | |
| 19. Universal | |
| 20. Wheels | |
| 21. Wheel Steer Parts (Electric) | |

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Copies of this standard for Military use may be obtained as indicated in the forward to or general provisions of the Index of Military Specifications and Standards.

Copies of this standard may be obtained for other than official use by individuals, firms and contractors from the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

Both the title and identifying symbol number should be stipulated when requesting copies of Military standards.

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